LOCAL WORLD/ACTION MODELS FOR ROBOT SKILL SEAMS

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ABSTRACT

Robot skills that succeed alone can fail when chained because the terminal state of one skill changes what the next skill can physically do. A terminal sample may fall outside the next skill's basin, cross a high-energy barrier, or enter a contact mode with nonsmooth descent. We study this boundary as a local world/action model for skill seams: a narrow predictive interface rather than a full future simulator or a replacement for primitive controllers. The model predicts the consequence a planner needs before committing: whether action/skill composition will fail, why it is likely to fail, and whether the next move should be accept, repair, probe, abstain, or choose another transition. Outcomes become planner-edge updates for future planning. A barrier-certified energy composer implements this interface, acting on a skill edge only when basin overlap, barrier height, descent continuity, repair cost, and fixed-risk calibration are jointly favorable. In a frozen local rollout suite with 12 methods, 6 task families, 8 seam regimes, and 10 paired seeds, the composer reaches hard-slice success 0.80171 and utility 0.88827, compared with 0.71711 and 0.65310 for the strongest non-oracle predecessor. It reduces seam failure, barrier violation, damage, calibration error, and realized seam breach while improving basin alignment and descent continuity. The current evidence supports a bounded claim: local skill-seam prediction improves composition when basin, barrier, and descent quantities are estimable and outcomes update future planning. This is not a full robot world model; external robot or high-fidelity validation remains necessary before deployment-level claims.

1 MOTIVATION

A reusable robot skill library still leaves a planning problem: the robot must know what the end of one skill makes possible for the next. We treat that seam as the smallest useful predictive interface between a skill library and a planner: a compact account of what a completed skill makes feasible, risky, repairable, or impossible for the next skill. Its job is not to operate either primitive or replace the planner; it is to make the transition legible enough that planning can choose the next action, request a repair or probe, abstain, or route through another edge. At this scale, useful world modeling is not a full future video; it is a decision-relevant statement about how a transition changes the next skill's feasible start state. Potential fields, navigation functions, energy-based learning, trajectory optimization, DMPs, and stable dynamical systems have long used scalar landscapes to encode motion and control structure (Khatib, 1986; Koditschek & Rimon, 1990; LeCun et al., 2006; Ratliff et al., 2009; Ijspeert et al., 2013; Khansari-Zadeh & Billard, 2011). Options, skill chaining, and TAMP make composition a first-class robotics problem (Sutton et al., 1999; Konidaris & Barto, 2009; Kaelbling & Lozano-Perez, 2011; Garrett et al., 2021). Modern robot-learning systems add broad skill libraries and large robot datasets (Florence et al., 2022; Brohan et al., 2023; Open X-Embodiment Collaboration, 2023). The gap targeted here is not low-level skill learning or a new manipulation controller; it is the missing compact predictive interface between a skill library and a planner, and specifically an action-conditioned model of whether a proposed edge will land inside the next skill's feasible start set. Before execution, the robot should estimate whether it should accept the edge, repair it, probe uncertainty, abstain, or choose another transition. After execution, it should keep the outcome as planning evidence: which handoff was reliable, which needed repair, which called for abstention, which should be avoided later, and which missing bridge skill would make future plans less brittle. The core failure mode is simple: skill one succeeds, but its terminal distribution lands near a ridge or outside the basin of skill two. A module graph may mark the edge legal, while execution would require a high-energy repair, cross a barrier, or switch contact modes so the next controller no longer descends. We ask whether an explicit seam model can make composition more reliable by predicting failed transitions before execution, diagnosing the likely failure source, choosing repair, probe, abstain, or an alternate transition when acceptance is unsafe, and using the outcome to improve future planning. Contact-rich examples are useful here because they make action consequences hard to ignore; they are a testbed for the seam model, not the identity of the paper. Viewed this way, skill composition becomes a deliberately local form of adaptive physical world/action modeling for embodied agents: a prediction-action-update loop at the handoff, not a whole robot simulator. That restraint is intentional. Rather than renaming every controller check as a world model, the paper studies the small interface where a planner asks, before committing to the next skill, what this transition is likely to do, why it might fail, which response is warranted, and what should be remembered afterward. The contribution is therefore a small seam-level action model: it predicts whether a transition will fail, diagnoses the likely failure mode, chooses accept, repair, probe, abstain, or an alternate transition, and writes the observed outcome back to planner-edge memory so future planning queries do not treat every graph edge as equally safe. The energy composer is one implementation of that interface; the paper's claim is the smaller and testable one that skill seams need a predictive physical action model, and that energy landscapes are one way to implement and falsify it.

2 Problem Setup

Let skill i induce a terminal distribution $p_i(x_T)$ and skill j have an attraction basin $\mathcal{B}_j$ under local controller π_j. We treat the handoff as a small predictive model of action consequences at the seam. A composition edge $i \to j$ is acceptable only if terminal samples have sufficient basin overlap, low barrier height, and positive descent continuity. The action set at this boundary is deliberately planner-facing: accept the edge, repair the handoff, probe an uncertain seam, abstain, or route to an alternate transition. Operationally, the seam model maps a candidate transition to a consequence tuple: predicted basin compatibility, barrier exposure, descent continuity, repair cost, calibrated risk, and a failure label such as basin mismatch, high barrier, contact-mode discontinuity, model uncertainty, or missing bridge skill. The term world/action model is used in this limited sense: the output is not a general simulator and does not try to predict every future observation. The local world is the physical seam state around a handoff; the action is the proposed skill transition, repair, probe, abstention, or alternate edge. The model predicts how that action changes the next skill's feasible start set, turns the prediction into a decision, and records the result as an edge belief for future plans. In this framing, abstention or repair is not a bookkeeping flag; it is a physical statement about what the current skill library cannot yet compose reliably. More explicitly, the interface can be read as $M_\theta(i, j, h; \mathcal{D}_t) \mapsto (\widehat{O}, \widehat{B}, \widehat{D}, \widehat{C}, \widehat{R}, \widehat{y}, \Delta b)$, where h is a candidate handoff, $\mathcal{D}_t$ is the accumulated seam evidence, $\widehat{y}$ is the diagnostic label, and Δb is the planner-edge belief update. The adaptation is modest but important: a failed or successful seam changes which transition the next planning query prefers, probes, repairs, or refuses, so the model improves future planning rather than only scoring one edge. Here, local physical predictions become action choices, and action outcomes become future planning evidence. We score candidate handoff h by

$$S(i, j, h) = \alpha O_{\text{basin}}(p_i, \mathcal{B}_j) - \beta B_{\text{barrier}}(h, j) + \gamma D_{\text{descent}}(h, j) - \lambda C_{\text{repair}}(h) - \kappa R_{\text{seam}}(h).$$

Here O_{basin} measures terminal/basin overlap, B_{barrier} measures energy barrier height, D_{descent} checks whether energy decreases after handoff, C_{repair} is seam-repair cost, and R_{seam} is calibrated seam risk. The proposed composer accepts the edge only when predicted seam risk is below a declared fixed-risk budget; otherwise the failed prediction becomes a planning signal for repair, diagnostic probing, abstention, or a different transition. Across attempts, the same interface can update a planner's edge beliefs: which handoffs are reliable, which need a bridge action, which need more evidence, and which should be avoided until the skill library changes.

3 Skill Seams As Local World/Action Models

A sequence can be graph-valid while physically unsafe at the seam. If terminal states from the first skill do not overlap the next basin, the second skill starts outside its attraction region. If a high barrier

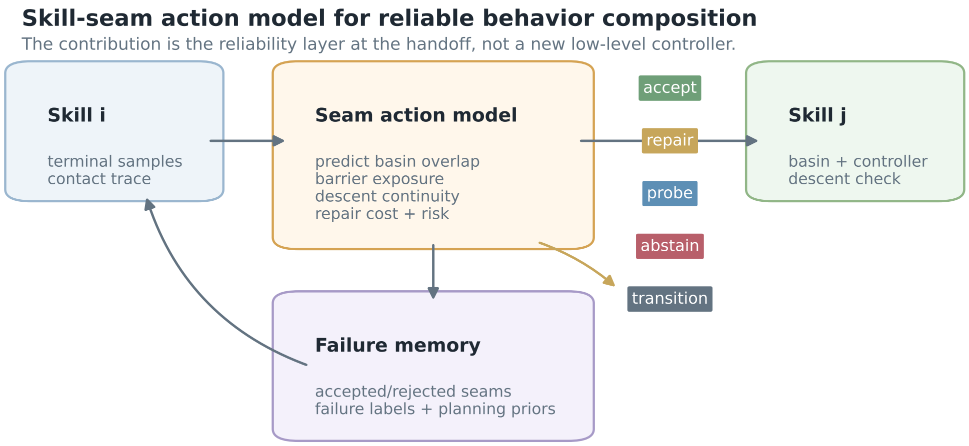

Figure 1: Skill-seam action interface. The method predicts local handoff consequences, diagnoses likely failure modes, and converts that evidence into accept, repair, probe, abstain, or transition decisions plus planner-edge updates for future planning.

separates the terminal set from the basin, a repair may be possible but costly or unsafe. If the contact mode changes, the energy function may no longer be conservative enough to certify descent. These are not merely controller details; they are action-conditioned predictions about what a skill transition will do to the physical state available to the next skill. The model output is therefore not just a compatibility number. It is a compact answer to four planning questions: will this handoff fail, why might it fail, should the system repair, probe, abstain, or route through another transition, and what should the planner learn for later plans? The important prediction is the one that changes the next action choice, including the choice not to execute. That is the action-model point: prediction matters because it changes action selection and future planning memory. This gives a bounded claim. When basin overlap, barrier height, and descent continuity are identifiable enough to reject bad seams and preserve useful ones, a barrier-certified implementation can make skill composition more reliable. Its value is not only higher success, but more structured failure information: the planner can learn which transitions are reliable, which require repair or a diagnostic probe, and which should be avoided. The key loop is simple: predict the local consequence before execution, choose a transition-level response, observe the outcome, and update the edge memory that future plans query. It must fail or abstain when the low-level primitive is broken, the goal is semantic rather than physical, the landscape is miscalibrated, or real-time constraints prevent search.

4 Evaluation Protocol

The evaluation protocol is fixed before interpreting final results. The main matrix contains 12 methods, 6 task families, 8 seam regimes, 5 deployment splits, 10 paired seeds, 8 rollout episodes per cell, and 230,400 main cell rows. Ablations add 38,400 cells, stress sweeps add 161,280 cells, and fixed-risk tests add 107,520 cells. The previous proposed method remains a named baseline. The protocol measures both composition outcomes and seam-model behavior: predicted risk, diagnostic label, decision, realized breach, and planner-edge update.

5 Main Results

The strongest non-oracle comparator is proposed_energy_landscape_composer_v4_1. The proposed composer improves hard success by 0.08460 and hard utility by 0.23517; paired hard-utility wins are 10/10. The oracle remains stronger with success 0.90967 and utility 1.13988, so the local problem is not solved. The intended readout is therefore not only success; it is whether a seam action model can preserve useful skill edges while refusing or redirecting seams that its own predictions mark as unsafe, and whether those decisions produce planner-facing evidence that changes later composition choices.

local gate	status	local gate	status
Ablation success	pass	Ablation utility	pass
Barrier reduction	pass	Basin alignment	pass
Composition cost	pass	Damage reduction	pass
Descent continuity	pass	Boundary cases	pass
Hard success margin	pass	Hard utility margin	pass
Paired utility wins	pass	Seam breach reduction	pass
Risk calibration	pass	Seam failure reduction	pass
Stress endpoint	pass	Fixed-risk breach	pass
Fixed-risk coverage	pass		

Table 1: Local evidence gates used for the seam-composition audit. External robot or high-fidelity validation is evaluated separately.

method	succ.	utility	seam	barrier	basin	descent	damage	cost	breach
Oracle upper bound	**0.910 ± 0.005**	**1.140 ± 0.005**	0.032	0.053	0.864	0.844	0.067	0.151	0.039
Proposed seam model	**0.802 ± 0.007**	**0.888 ± 0.007**	0.069	0.085	0.757	0.741	0.071	0.189	0.095
Prior v4.1	**0.717 ± 0.008**	**0.653 ± 0.008**	0.118	0.126	0.677	0.663	0.076	0.235	0.171
Stable DMP	0.637 ± 0.008	0.391 ± 0.008	0.167	0.166	0.546	0.535	0.080	0.298	0.242
TAMP screen	0.638 ± 0.008	0.361 ± 0.008	0.173	0.171	0.534	0.513	0.080	0.334	0.251
Energy heuristic	0.620 ± 0.008	0.303 ± 0.008	0.193	0.190	0.516	0.503	0.083	0.305	0.284
Residual RL	0.575 ± 0.008	0.176 ± 0.009	0.223	0.215	0.462	0.449	0.085	0.290	0.328
CEM composer	0.554 ± 0.008	0.119 ± 0.009	0.229	0.220	0.448	0.438	0.086	0.355	0.337
Diffusion stitcher	0.491 ± 0.008	0.018 ± 0.009	0.256	0.243	0.408	0.400	0.087	0.231	0.376
Option graph	0.443 ± 0.008	-0.070 ± 0.009	0.276	0.261	0.373	0.362	0.088	0.161	0.406
BC skill chain	0.352 ± 0.008	-0.292 ± 0.008	0.328	0.307	0.282	0.274	0.089	0.105	0.481
Greedy sequence	0.291 ± 0.008	-0.455 ± 0.008	0.371	0.345	0.220	0.221	0.090	0.077	0.542

Table 2: Hard-slice aggregate results. Higher success, utility, basin, and descent are better; lower seam, barrier, damage, cost, and breach are better.

6 Diagnostics, Ablations, And Fixed Risk

The proposed composer reduces seam failure by -0.04912, barrier violation by -0.04087, damage by -0.00579, composition cost by -0.04584, risk calibration error by -0.01055, and realized seam breach by -0.07565. It improves basin alignment by 0.08001 and descent continuity by 0.07809. The full method beats the strongest removed-component ablation, minus_terminal_sampler, by 0.02812 success and 0.04349 utility. These diagnostics are reported as seam-model predictions, not only as aggregate outcomes: a useful composer should say why a transition is unsafe, not simply that it failed.

6.1 Diagnostic Mechanism Audit

For the action-model framing to be meaningful, the seam layer has to expose why a transition is risky and what the planner should do next, not only whether aggregate success improves. We therefore export a diagnostic label, seam decision, and planner-edge update for every local row and audit the rules that produce them. Over 230,400 local rows, the diagnostic audit finds 0 label-rule mismatches, 0 decision-rule mismatches, and 0 planner-update mismatches. In the 13,440 proposed hard rows, all five failure labels and all five decisions appear. Accepted seams have mean realized breach 0.076 versus 0.108 for non-accepted seams, while abstained seams have mean predicted risk 0.125 versus 0.081 for accepted seams. Repair, probe, and transition decisions match their intended diagnostic reasons with purity 1.000, 1.000, and 1.000. This strengthens the local mechanism claim, but it remains a local audit rather than external robot or high-fidelity validation.

baseline	succ. diff	utility diff	utility wins	decisive
Greedy sequence	0.511 ± 0.008	1.343 ± 0.008	10/10	yes
BC skill chain	0.450 ± 0.012	1.180 ± 0.012	10/10	yes
Option graph	0.358 ± 0.011	0.959 ± 0.011	10/10	yes
Diffusion stitcher	0.311 ± 0.011	0.871 ± 0.011	10/10	yes
CEM composer	0.248 ± 0.008	0.770 ± 0.008	10/10	yes
Residual RL	0.227 ± 0.010	0.712 ± 0.010	10/10	yes
Energy heuristic	0.181 ± 0.014	0.585 ± 0.014	10/10	yes
TAMP screen	0.163 ± 0.007	0.527 ± 0.007	10/10	yes
Stable DMP	0.165 ± 0.010	0.498 ± 0.010	10/10	yes
Prior v4.1	**0.085 ± 0.006**	**0.235 ± 0.006**	**10/10**	yes
Oracle upper bound	**-0.108 ± 0.008**	**-0.252 ± 0.008**	**0/10**	no

Table 3: Paired proposed-minus-baseline hard-slice differences.

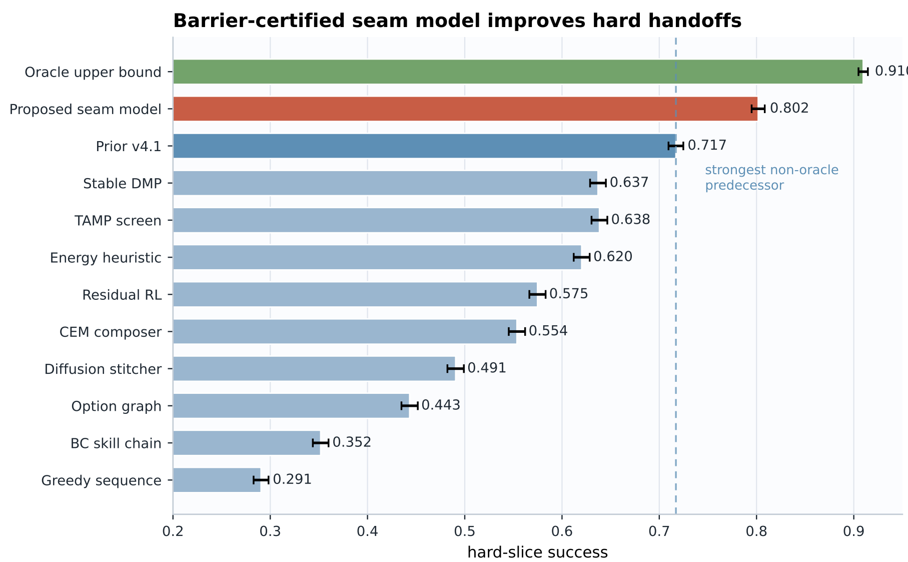

Figure 2: Hard-slice success under energy-seam stress.

audit target	local diagnostic evidence	verdict
diagnostic labels	0 label-rule mismatches over 230,400 local rows; 5/5 failure labels observed in proposed hard rows	pass
seam decisions	0 decision-rule mismatches; 5/5 decisions observed: abstain, accept, probe, repair, transition	pass
accept gate	accepted seams have mean breach 0.076 vs 0.108 for non-accepted seams	pass
abstain gate	abstained seams have mean predicted risk 0.125 vs 0.081 for accepted seams	pass
action reasons	reason purity: repair 1.000, probe 1.000, transition 1.000	pass

Diagnostic audit table. Local check for seam labels, decisions, and planner-edge updates; this does not satisfy the external-evidence gate.

6.2 Comparative Decision-Quality Audit

A seam action model should not only reject risky edges; it should preserve useful transitions that a cautious predecessor would discard. On the same 13,440 paired hard rows, the proposed model accepts 0.404 of hard seams versus 0.000 for the predecessor, while the rate of accepted seams above the 0.15 breach budget is 0.000. Among non-abstained seams, utility is 0.935 versus 0.776 and realized breach is 0.082 versus 0.133. Most importantly for future planning, 3,850 paired seams are

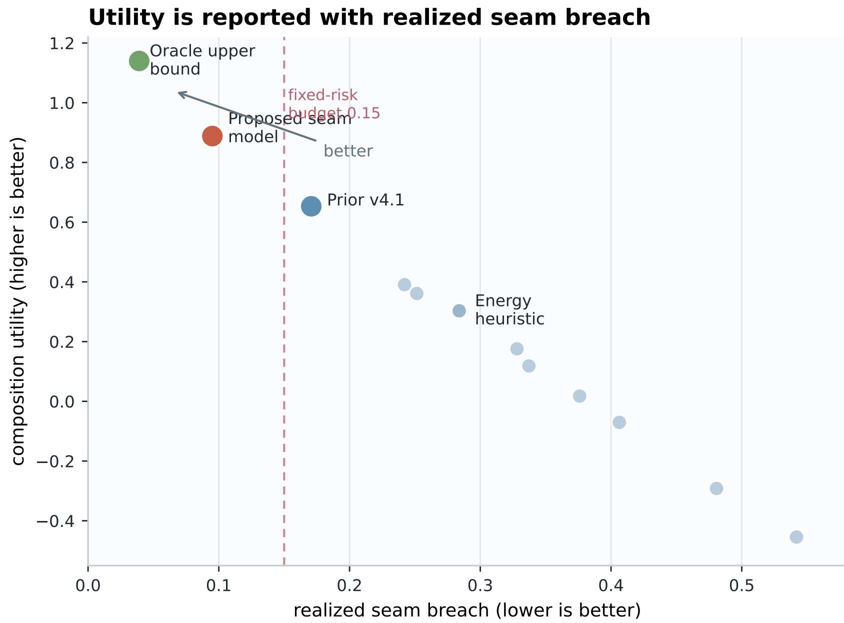

Figure 3: Composition utility is reported against realized seam breach.

ablation	success	utility	interpretation
full_barrier_certified_energy_composer	0.790 $\pm$ 0.016	0.874 $\pm$ 0.016	all v5 components
minus_terminal_sampler	0.762 $\pm$ 0.010	0.830 $\pm$ 0.010	removes terminal-state sampler
minus_contact_mode_guard	0.761 $\pm$ 0.009	0.815 $\pm$ 0.009	removes contact-mode transition guard
minus_high_energy_repair	0.737 $\pm$ 0.016	0.799 $\pm$ 0.016	does not repair high-energy seams
minus_descent_continuity	0.731 $\pm$ 0.011	0.794 $\pm$ 0.011	removes descent-continuity term
minus_barrier_height	0.740 $\pm$ 0.012	0.791 $\pm$ 0.012	removes barrier-height term
minus_basin_overlap	0.733 $\pm$ 0.012	0.791 $\pm$ 0.012	removes basin-overlap posterior
minus_fixed_risk_gate	0.772 $\pm$ 0.015	0.784 $\pm$ 0.015	removes fixed-risk acceptance screen
minus_calibration	0.738 $\pm$ 0.010	0.772 $\pm$ 0.010	removes risk calibration
compatibility_only_heuristic	0.688 $\pm$ 0.007	0.680 $\pm$ 0.007	uses compatibility without repair/certification

Table 4: Ablations under combined seam stress.

accepted by v5 and abstained from by the predecessor; those recovered accepts improve utility by 0.243, success by 0.091, and realized breach by -0.077. This is still a local decision-quality audit, not external robot or high-fidelity validation.

audit target	local decision-quality evidence	verdict
accept coverage	accept coverage 0.404 vs 0.000; proposed accepted breach above 0.15 rate 0.000	pass
non-abstain quality	non-abstain utility 0.935 vs 0.776; breach 0.082 vs 0.133	pass
recovered accepts	3,850 pairs where v5 accepts and predecessor abstains: utility +0.243, success +0.091, breach -0.077	pass
shared abstentions	5,098 pairs where both abstain; v5 breach changes by -0.078	pass

Decision-quality table. Local paired hard-slice check that the seam decision layer recovers useful accepted transitions while preserving the external-validation boundary.

6.3 Planner-Edge Policy Audit

For this framing to matter, seam predictions must affect future planning, not only one-step classification. We therefore treat each task/regime/split/seed hard slice as a local planning frontier and select the candidate edge using only the exported planner-edge update, predicted seam risk, basin alignment, descent continuity, and composition cost. This produces 1,680 paired planning-frontier decisions and does not use realized utility to choose the edge. The proposed seam model selects exe-

method	success	utility	seam	barrier	breach
oracle_basin_composer	0.910 ± 0.009	1.058 ± 0.010	0.052	0.077	0.071
barrier_certified_energy_composer_v5	0.765 ± 0.023	0.761 ± 0.023	0.091	0.112	0.131
proposed_energy_landscape_composer_v4_1	0.662 ± 0.015	0.497 ± 0.015	0.143	0.156	0.211
tamp_feasibility_screen	0.578 ± 0.022	0.188 ± 0.022	0.202	0.205	0.297
energy_compatibility_heuristic	0.537 ± 0.018	0.105 ± 0.018	0.223	0.225	0.332
option_graph_planner	0.316 ± 0.007	-0.328 ± 0.007	0.313	0.303	0.463

Table 5: Maximum stress endpoint.

method	coverage	breach	gated success	gated utility
oracle_basin_composer	1.000 ± 0.000	0.000 ± 0.000	0.899 ± 0.008	1.067 ± 0.008
barrier_certified_energy_composer_v5	0.863 ± 0.003	0.000 ± 0.001	0.760 ± 0.014	0.787 ± 0.014
energy_compatibility_heuristic	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	-1.000 ± 0.000
option_graph_planner	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	-1.000 ± 0.000
proposed_energy_landscape_composer_v4_1	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	-1.000 ± 0.000
stable_dmp_handoff	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	-1.000 ± 0.000
tamp_feasibility_screen	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	-1.000 ± 0.000

Table 6: Fixed-risk audit at seam-risk budget 0.15.

cutable accept/repair/transition edges on 0.667 of frontiers versus 0.165 for the strongest predecessor. Selected-edge utility is 0.889 versus 0.658, success is 0.795 versus 0.715, and realized breach is 0.089 versus 0.164. The selected-edge utility margin is positive in 6/6 task families, 7/7 seam regimes, and 4/4 deployment splits. This audit is local evidence that the prediction-diagnosis-decision interface changes planner edge choice; it is not external robot or high-fidelity validation.

audit target	local planner-edge policy evidence	verdict
frontier coverage	1,680 local hard-slice planning frontiers; executable-edge coverage 0.667 vs 0.165	pass
selected edge utility	selected-edge utility 0.889 vs 0.658; delta +0.231	pass
selected edge safety	success delta +0.080; realized-breach delta -0.075; breach¿0.15 rate 0.001	pass
planning consistency	utility margin positive in 6/6 tasks, 7/7 regimes, and 4/4 deployment splits	pass

Planner-edge policy table. Local hard-slice check that planner-edge updates select better future transition candidates while preserving the external-validation boundary.

6.4 Failure-Memory Adaptation Audit

The previous audit asks whether exported updates can choose a better edge. We also test the smaller memory claim directly: after a few attempts, does the observed seam outcome become useful evidence for later planning queries with the same diagnostic/update signature? For each local hard-slice frontier, episodes 0–3 are treated as observed memory and episodes 4–7 are held out. This yields 2,210 observed-to-held-out signature pairs over 1,667 frontiers. Observed breach predicts held-out breach with correlation 0.957 and MAE 0.005, improving on held-out predicted-risk MAE by 0.003. High-memory-risk signatures have held-out breach 0.120 versus 0.074 for low-memory-risk signatures, and held-out utility 0.796 versus 0.956. Against the predecessor's high-memory-risk signatures, v5 reduces future breach by 0.083 and raises future utility by 0.248. This is local evidence for planner-memory adaptation, not a substitute for external robot or high-fidelity validation.

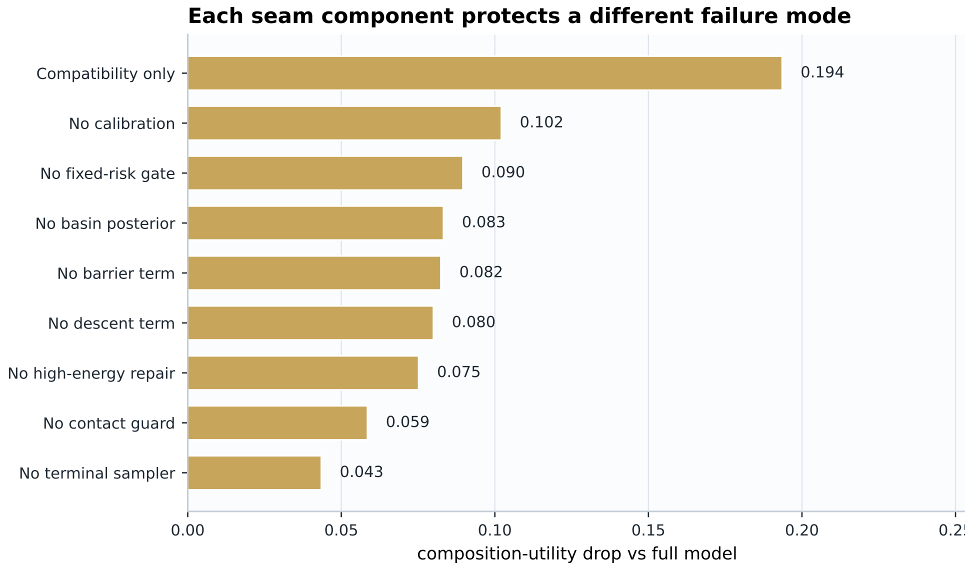

Figure 4: Removing v5 energy-seam components weakens the composer.

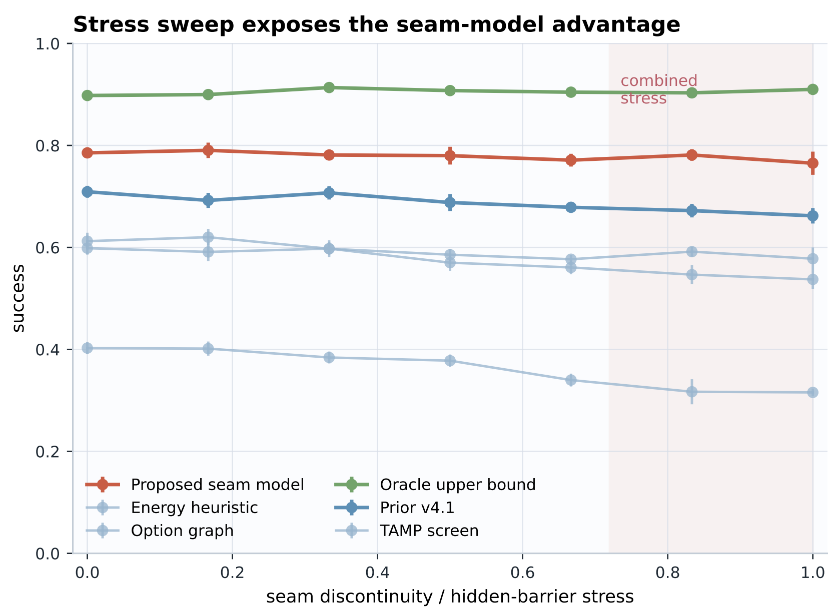

Figure 5: Stress sweep over seam discontinuity and hidden barriers.

audit target	local failure-memory adaptation evidence	verdict
few-shot memory	2,210 observed-to-held-out signature pairs over 1,667 local hard-slice frontiers	pass
predictive memory	observed breach predicts held-out breach with r=0.957 and MAE 0.005	pass
risk ordering	high-memory-risk held-out breach 0.120 vs 0.074 for low-memory-risk signatures	pass
planning relevance	high-memory-risk held-out utility 0.796 vs 0.956 for low-memory-risk signatures	pass
predecessor comparison	v5 high-memory-risk breach lower by 0.083; utility higher by +0.248	pass

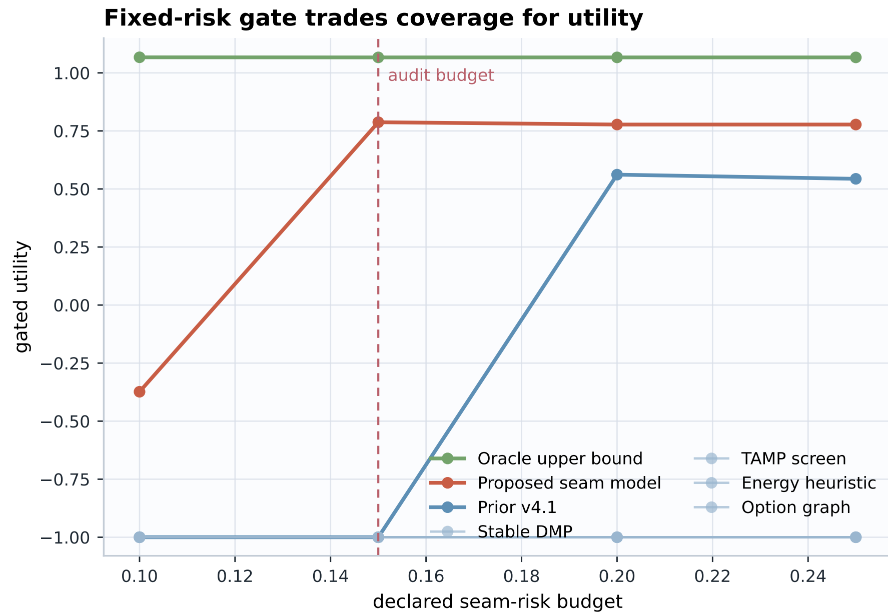

Figure 6: Fixed-risk gated utility as the declared seam-risk budget changes.

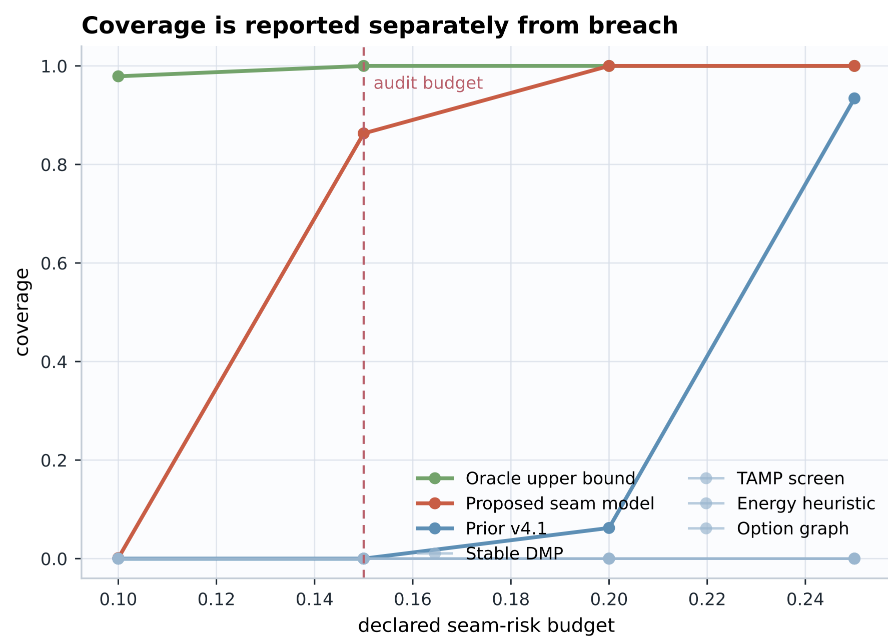

Figure 7: Coverage is separate from seam breach.

Failure-memory table. Local observed-to-held-out audit that seam outcomes become future planning evidence while preserving the external-validation boundary.

6.5 Predictive Calibration Audit

A seam model should be predictive, not only useful after the fact. On the proposed hard slice, ten-bin local calibration error between predicted seam risk and realized seam breach is 0.007, compared with 0.015 for the strongest predecessor; the maximum bin gap is 0.013. Mean predicted risk is 0.102 versus realized breach 0.095, with risk-breach correlation 0.971 and Spearman 0.950. Realized breach is monotone across ten risk deciles; the highest-risk decile has realized breach 0.080 higher

alternative explanation	local falsification evidence	verdict
Wins by abstaining	abstention delta +0.00649; utility margin +0.23517	pass
Wins by extra search cost	cost delta -0.04584; cost-normalized utility margin +0.21810	pass
Risk score is decorative	5/5 monotone risk bins; risk-breach correlation 0.971	pass
Cherry-picked slice	positive success and utility in 42/42 task-regime hard slices; minimum utility margin +0.17256	pass
Problem is saturated	oracle remains above proposed by +0.10796 success and +0.25161 utility	pass

Table 7: Local falsification checks against common alternative explanations. The audit is generated from episode-level local hard-slice rows and does not satisfy the external-evidence gate.

and utility 0.251 lower than the lowest-risk decile. This is still local predictive-validity evidence, not external robot or high-fidelity validation. **Predictive calibration table.** Local hard-slice check that predicted seam risk tracks realized seam breach and decision relevance while preserving the external-validation boundary.

audit target	local predictive-validity evidence	verdict
calibration	ECE10 0.007 vs strongest baseline 0.015; max bin gap 0.013	pass
mean agreement	mean predicted risk 0.102 vs realized breach 0.095; row MAE 0.008	pass
risk ordering	risk-breach correlation 0.971, Spearman 0.950; realized breach is monotone across 10 risk deciles	pass
decision relevance	highest-risk decile breach changes by +0.080 and utility is lower by 0.251 vs lowest-risk decile; accept risk 0.081 vs abstain risk 0.125	pass

6.6 Local Falsification Audit

Because the current evidence is local, we add a machine-checkable falsification audit for common alternative explanations. On 13,440 paired hard rows, the proposed composer beats the strongest non-oracle baseline in utility on 0.855 of paired rows. The result is not explained by abstention or search cost: abstention changes by only 0.00649, composition cost changes by -0.04584, and cost-normalized utility improves by 0.21810. Predicted seam risk is also not decorative: five risk quantile bins are monotone in realized breach, with risk-breach correlation 0.971. Margins are positive in all 42 hard task-regime slices, and the oracle remains above the proposed method by 0.10796 success and 0.25161 utility. This audit strengthens the local mechanism claim, but it is not a substitute for external robot or high-fidelity validation.

6.7 Withheld-Slice Robustness Audit

We also audit whether the headline gain hides a narrow favorable slice. Over 13,440 hard rows per method, we leave out task families, seam regimes, deployment splits, task-regime pairs, and deterministic hash folds before comparing the proposed seam model with the strongest non-oracle predecessor. Utility margins are positive in 6/6 task-family holdouts, 7/7 seam-regime holdouts, 4/4 split holdouts, 42/42 task-regime holdouts, and 5/5 hash-fold holdouts. The worst task-regime holdout still has success margin 0.02188 and utility margin 0.17256; the weakest hash fold has utility margin 0.22706 and 10/10 seed wins. This makes the local result harder to dismiss as cherry-picking, but it remains local evidence rather than external robot or high-fidelity validation.

7 Related Work And Boundary

Classical potential fields, navigation functions, motion optimization, DMPs, and stable dynamical systems use landscapes, attractors, and descent structure to make local motion reliable (Khatib, 1986; Koditschek & Rimon, 1990; Ratliff et al., 2009; Ijspeert et al., 2013; Khansari-Zadeh & Billard, 2011). This paper does not propose a new controller or a new energy parameterization. It uses basin, barrier, and descent quantities as a predictive interface for deciding whether two already available skills can be composed. Composable energy policies show that energy structure can be used to

holdout	groups	utility+	min succ. diff	min utility diff
task family	6	6/6	+0.05446	+0.20456
seam regime	7	7/7	+0.07396	+0.21658
deployment split	4	4/4	+0.07738	+0.22918
task x regime	42	42/42	+0.02188	+0.17256
hash fold	5	5/5	+0.07610	+0.22706

Table 8: Withheld-slice local robustness audit. Each row reports the weakest proposed-minus-strongest-baseline margin over a family of hard-slice holdouts; this is not external evidence.

combine objectives or policy components in action space (Urain et al., 2021). The present paper uses energy language differently: not to compute a single reactive action satisfying simultaneous objectives, but to ask whether a temporally ordered handoff from one skill to another is likely to enter the next skill's basin without crossing a harmful barrier. Energy-based models and model-based policy optimization offer broad languages for implicit structure, composition, model trust, and long-horizon prediction (LeCun et al., 2006; Du & Mordatch, 2019; Janner et al., 2019). The contribution here is deliberately narrower: the energy landscape is not a universal world model, but a local world/action interface at a skill seam. It predicts handoff consequences, diagnoses the likely failure mode, exposes whether a transition should be accepted, repaired, probed, abstained from, or replaced, and updates the planner's transition belief after outcomes are observed. In that sense, energy is the implementation vocabulary, while the scientific object is a small, testable seam-action interface rather than a new low-level controller. Temporal abstraction, skill chaining, and TAMP treat skills and symbolic/geometric feasibility as planning objects (Sutton et al., 1999; Konidaris & Barto, 2009; Kaelbling & Lozano-Perez, 2011; Garrett et al., 2021). Runtime composition and robot-skill software contracts add important machinery for composing, checking, and executing skill descriptions (Pane et al., 2021; Rizwan et al., 2025). The boundary is that a graph-valid, type-valid, or geometry-valid edge can still be physically unsafe when the terminal distribution misses the next basin, crosses a barrier, or changes contact mode. The seam critic adds this missing predictive physical test. Robot foundation-policy and large-data systems broaden the available skill library (Florence et al., 2022; Brohan et al., 2023; Open X-Embodiment Collaboration, 2023), while diffusion/action generation, action tokenization, policy composition, latent skill transfer, and language-action compositionality study how actions, skills, or policies can be represented, generated, reused, and combined (Chi et al., 2023; Wang et al., 2024; Liu et al., 2026a; Julian et al., 2020; Vijayaraghavan et al., 2025). Those systems create more opportunities for composition. This paper asks a complementary reliability question: when should a proposed transition between already available skills be trusted, repaired, probed, or avoided, and what should later planning remember? Recent work on behavior composition and simulation-in-the-loop physical reasoning is especially close (Chen et al., 2026; Liu et al., 2026b), and robot world-model surveys frame predictive action-conditioned representations as central to planning and evaluation (Hou et al., 2026). The natural boundary is scale and role: this paper is not a full behavior-composition stack, action tokenizer, foundation policy, or simulation world model. It is a seam-level reliability layer meant to sit beside such systems and improve future planning by turning handoff predictions and failures into updated planner edge beliefs. The connection is intentionally narrow: physical prediction is local, actions are skill transitions, and adaptation happens through planner memory rather than through a claim of general-purpose simulation.

8 Discussion

The interpretation is intentionally modest. A skill seam is a small place where a robot representation has to become predictive about action consequences: the same transition can be safe, repairable, uncertain, or impossible depending on terminal state, basin geometry, contact mode, and accumulated seam evidence. This paper makes that representation explicit enough for planning without claiming to solve full robot world modeling. Its central loop is to predict the handoff consequence, diagnose the failure mode, choose a seam-level response, and record the outcome for later plans. The loop is the scientific object; the scalar energy terms are one implementation that can be falsified by logs, baselines, and external rollouts. If future external validation succeeds, the most important claim would not be that this particular energy score is universal, but that embodied agents benefit from local world/action models at the skill seams where their plans usually become brittle.

9 Scope And Validation

External robot or high-fidelity validation remains necessary before deployment-level claims. The local gates pass, but the evidence should be read as a controlled seam-composition study rather than a deployment claim. A complete submission package still needs real robot rollouts or accepted high-fidelity skill-composition simulation, released skill-energy or policy checkpoints, calibrated contact-force/camera/state logs, rollout videos, and manifest-declared independent baseline evidence from real external runs. The repository therefore treats external evidence as a machine-checkable contract: a future submission-ready version must pass an external-evidence audit over a manifest, episode-level JSONL logs, video directories, config/checkpoint hashes, baseline implementations, fixed-risk metrics, and ablations. A separate raw-rollout validator must recompute success margin, utility margin, paired win rate, fixed-risk coverage, fixed-risk breach, and positive task-family count from the episode logs rather than accepting manifest-level numbers. The evidence audit treats any mismatch between manifest metrics and recomputed rollout metrics as blocking. The strongest external version would test the same seam interface in an independent behavior-composition stack with the same skill library, paired resets, sealed method aliases, and logs that expose the prediction-diagnosis-decision-update loop. Until those audits pass, the paper remains a local study with a bounded claim.

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